

Machine Learning for teens

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SVM

► Support Vector Machines (SVM)

- **Support Vector Machine** is a supervised learning algorithm used for **classification** and sometimes **regression**.

- **Margin**

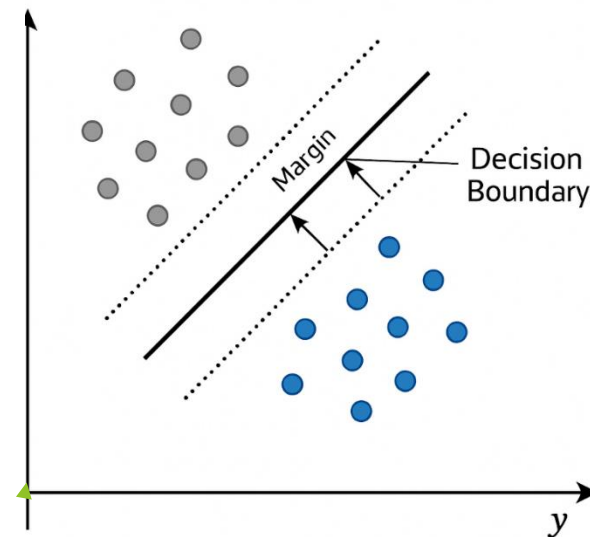
- the space between the decision boundary and the closest data points from each class

- **support vectors**

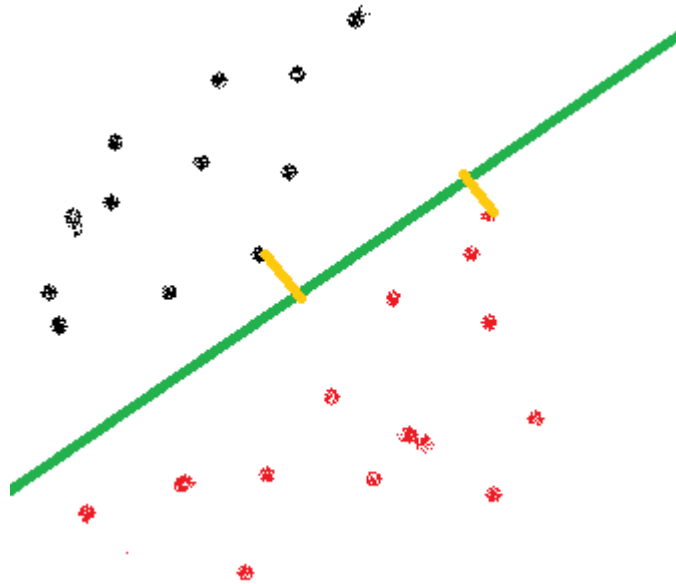
- These closest points are called **support vectors**

- SVM chooses the boundary that **maximizes this margin**

- Get the code from [here](#)



Better visualization for SVM



Ways to Overcome Overfitting

- ▶ **Feature Selection**
 - ▶ Simplify the model by removing irrelevant or noisy features.
- ▶ **Regularization**
 - ▶ Regularization works for linear models.
- ▶ **Cross-Validation**
 - ▶ KFold, StratifiedKFold, cross_val_score
- ▶ **Model Constraints** (for Trees)
 - ▶ Like our code
- ▶ **Dimensionality Reduction**
 - ▶ Reduce feature space like PCA

Ways to Overcome Overfitting

- ▶ **Dropout**
 - ▶ For neural networks
- ▶ **Data Augmentation (for images/text)**
 - ▶ Increase training diversity.
 - ▶ Flip, rotate, crop images
 - ▶ Synonym replacement for text
- ▶ **Ensemble Methods**
 - ▶ Combine models to reduce variance.
 - ▶ Bagging (Random Forest), Boosting (XGBoost), Stacking

Dimensionality Reduction

▶ Principal Component Analysis (PCA)

- ▶ is a technique for **Dimensionality Reduction**
 - ▶ it **builds new features** called **principal components**. Each component is a **combination of all the original features**, with different weights.
- ▶ PCA is an **unsupervised** method
 - ▶ It **does not use the target label** at all
- ▶ Scale user data before using PCA
 - ▶ because PCA is sensitive to the **magnitude of feature values**
- ▶ Get the code from [here](#)

Assignment

- ▶ Use the Concrete Compressive Strength dataset (from `sklearn.datasets.fetch_openml`).
- ▶ Train an SVM model and check if it shows signs of overfitting or underfitting.
- ▶ Then apply PCA with different variance thresholds.
- ▶ Train the SVM model again on the PCA-transformed data and compare results.